that this asset class is perceived as being so low in risk, how much credit risk do munis really have?

Municipalities enter bankruptcy all the time. Orange County, California, caused a huge stir when it entered bankruptcy in 1994. In 2013, the City of Detroit filed for bankruptcy—the largest municipal bankruptcy in history with debts close to $20 billion. Since the financial crisis, scattered defaults among municipalities have arisen partly from the fall in house prices and the legacy of the subprime crisis. A partial list is as follows: in 2008, City of Vallejo, California; in 2009, City of Prichard, Alabama; in 2011, City of Central Falls, Rhode Island, City of Harrisburg, Pennsylvania, Jefferson County, Alabama; and in 2012, City of Stockton, California. But, overall, credit risk in munis is extremely low. The decomposition estimated by Ang, Bhansali, and Xing (2012) in equation (12.7) shows that credit risk plays a negligible part in municipal bond spreads, but this is for the sample used by Ang, Bhansali, and Xing since the mid-1990s. Perhaps muni risk now is higher?

Moody's reports only four defaults among towns, cities, or counties from 1970 to 2009, and only fifty-four defaults of municipal entities over the same period.[16] The average five-year cumulative default for investment grade muni debt is 0.03% compared to 1% for investment-grade corporate. Thus, credit risk in munis has been very low—an order of magnitude lower than credit risk for investment grade corporate debt. Moody's also reports that the recovery rates when a default occurred were 60 cents on the dollar, compared to 38 cents for investment grade corporate debt. Hempel's (1971) study gave special attention to the many municipal defaults that occurred during the 1920s and 1930s and also found that default rates were much lower, and recovery rates much higher, than for corporate debt.

Whitney's prediction may yet come true. States have gone bankrupt before.[17] In the early 1830s and 1840s, eight states and one territory defaulted: Arkansas, Florida Territory, Indiana, Louisiana, Maryland, Michigan, Mississippi, and Pennsylvania. U.S. states during that era were the emerging markets of the time, and there was heavy borrowing for infrastructure that could not be paid back ("canals to nowhere," in the words of John Cochrane).[18] Some states, like Florida and Mississippi, repudiated completely. In Charles Dickens's 1843 novel, *A Christmas Carol*, the protagonist Ebenezer Scrooge has a nightmare in which his wealth has withered into a "mere United States security." Scrooge was not

[16] U.S. Municipal Bond Defaults and Recoveries, 1970–2009, Moody's Investors Service, Feb. 2010.

[17] For more details see Ang and Longstaff (2013), who also compare the credit risk of U.S. states with the credit risk of Eurozone nations. U.S. states are sovereign borrowers in a currency union, just as Eurozone nations are. Germany, however, is not going to send troops to prevent a country from leaving the Eurozone, while the United States had Gettysburg.

[18] http://johnhcochrane.blogspot.com/2012/02/sargent-on-debt-and-defaults.html